Paper Link

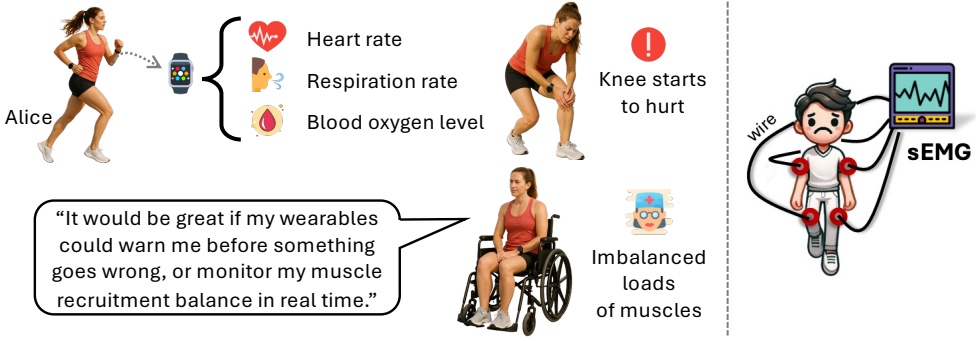
Exploring the Feasibility of Full-Body Muscle Activation Sensing with Insole Pressure Sensors

Hao Zhou, Mahanth Gowda

Motivation

Knowing Muscle Usage in Daily Life Could Provide More Health Insights

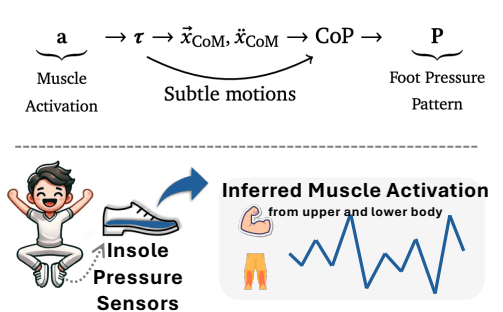
Current sensing methods require skin-mounted electrodes or cameras

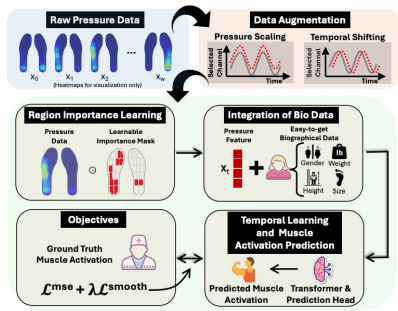


Our Solution

Press2Muscle: Learning Muscle Activation from Foot Pressure Signals

Leveraging biomechanical connections to infer muscle activation from foot pressure



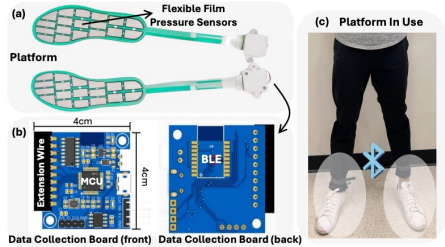


Platform & Evaluation

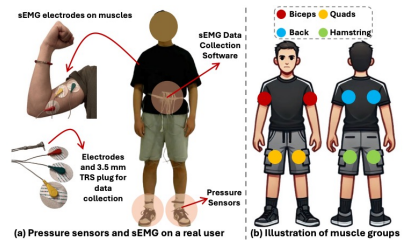
Press2Muscle is Evaluated on A Diverse Cohort

30 Participants, 15 Motions, 8 Muscle Groups, 2 Case Studies

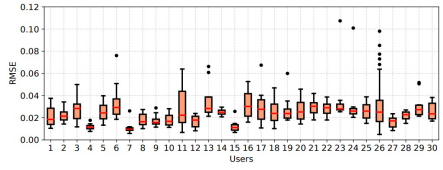
(a) Platform and Setup



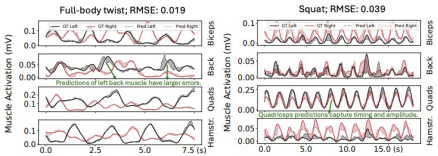
(b) Ground Truth Collection



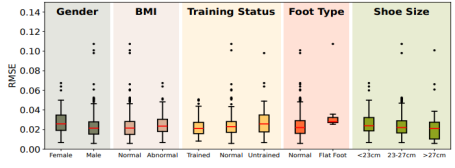
(c) Generalized to Unseen Users



(d) Qualitative Results



(e) Robustness



(f) Muscle Usage Imbalance Free-living

